EXPERIMENT – 1

AIM:

Set up Python environment for AI and implement a simple reflex or goal-based agent in a toy environment.

OBJECTIVE:

To set up a working Python environment for Artificial Intelligence programming and to design and implement a Simple Reflex Agent that senses its environment through percepts and acts using condition– action rules, using the classic Vacuum Cleaner World as the toy environment.

PROCEDURE:

1. Install Python 3 and set up the working environment (VS Code / Jupyter / Google Colab).
2. Define the toy environment — a two-room Vacuum World (Room A and Room B), each of which can be in a 'Dirty' or 'Clean' state.
3. Design the agent's percept, i.e., the current location and the status (dirty/clean) of that location.
4. Define condition–action (if–then) rules for the Simple Reflex Agent: if the current room is dirty, suck; else move to the other room.
5. Implement the VacuumEnvironment class to hold and display environment state.
6. Implement the SimpleReflexVacuumAgent class that perceives the environment and reflexively selects an action.
7. Run the agent for a fixed number of steps and observe how it cleans both rooms and stabilizes once the environment is fully clean.
8. Record the performance measure (a reward-based score) and the final state of the environment.

PROGRAM / CODE:

```
# ================================================================
 # PRACTICAL 1
 # Simple Reflex Agent - Vacuum Cleaner World
 # ================================================================
 class VacuumEnvironment:
     def __init__(self, status=None):
         self.locations = ["A", "B"]
         self.status = status if status else {"A": "Dirty", "B": "Dirty"}
       def display(self):
           print(f"Room A: {self.status['A']:<6} | Room B: {self.status['B']:<6}")
 class SimpleReflexVacuumAgent:
     def __init__(self):
         self.location = "A"
         self.performance = 0
     def act(self, environment):
         percept = environment.status[self.location]
          print(f"Agent at {self.location} | Percept: {percept}")
          if percept == "Dirty":
              print(f"Action : SUCK (cleaning Room {self.location})")
              environment.status[self.location] = "Clean"
              self.performance += 10
          elif self.location == "A":
              print("Action : MOVE RIGHT")
              self.location = "B"
              self.performance -= 1
          elif self.location == "B":
              print("Action : MOVE LEFT")
              self.location = "A"
```


Name: C.Jagadesh **Enrollment No: 92400118512**

```
                self.performance -= 1
    if __name__ == "__main__":
        env = VacuumEnvironment()
        agent = SimpleReflexVacuumAgent()
        print("=" * 55)
        print("      SIMPLE REFLEX AGENT - VACUUM CLEANER WORLD")
        print("=" * 55)
        print("\nInitial Environment State:")
        env.display()
        print("-" * 55)
        steps = 6
        for step in range(1, steps + 1):
            print(f"\nStep {step}")
            agent.act(env)
            env.display()
        print("-" * 55)
        print(f"Total Performance Score : {agent.performance}")
        print("Final Environment State :")
        env.display()
        print("=" * 55)
```

OUTPUT:

```
=======================================================
       SIMPLE REFLEX AGENT - VACUUM CLEANER WORLD
 =======================================================
 Initial Environment State:
 Room A: Dirty | Room B: Dirty
 -------------------------------------------------------
 Step 1
 Agent at A | Percept: Dirty
 Action : SUCK (cleaning Room A)
 Room A: Clean | Room B: Dirty
 Step 2
 Agent at A | Percept: Clean
 Action : MOVE RIGHT
 Room A: Clean | Room B: Dirty
 Step 3
 Agent at B | Percept: Dirty
 Action : SUCK (cleaning Room B)
 Room A: Clean | Room B: Clean
 Step 4
 Agent at B | Percept: Clean
 Action : MOVE LEFT
 Room A: Clean | Room B: Clean
 Step 5
 Agent at A | Percept: Clean
 Action : MOVE RIGHT
 Room A: Clean | Room B: Clean
 Step 6
 Agent at B | Percept: Clean
 Action : MOVE LEFT
 Room A: Clean | Room B: Clean
 -------------------------------------------------------
 Total Performance Score : 16
 Final Environment State :
 Room A: Clean | Room B: Clean
 =======================================================
```

RESULT:

The Python AI environment was configured successfully, and a Simple Reflex Agent was implemented for the two-room Vacuum Cleaner World. Using only current percepts and fixed rules, the agent cleaned both rooms successfully, demonstrating rational behaviour without memory or planning.